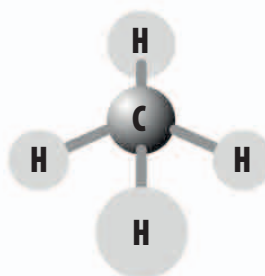


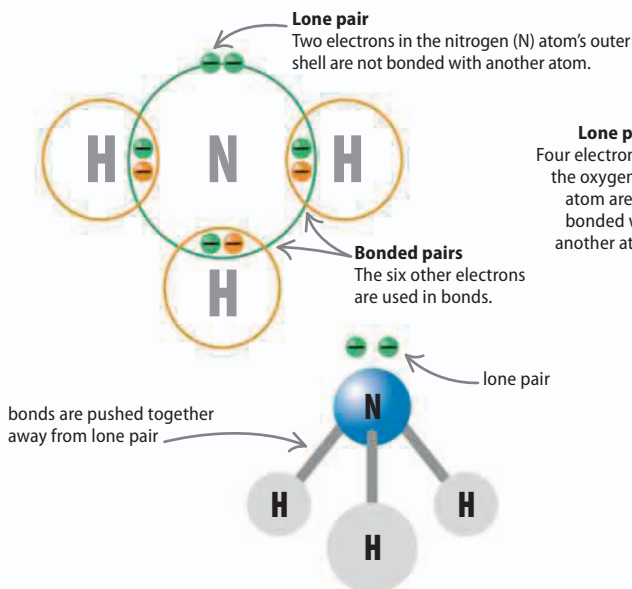
Shapes and bonds

In a methane molecule (see page 114), every electron in the carbon atom's outer shell is shared with a hydrogen atom. However, in other molecules not all the electrons are involved in a bond. The ones that are not are called lone pairs. These lone pairs create a zone of electric charge that repels the bonded pairs and pushes them closer together. So the arrangement of bonded and lone pairs gives a molecule its shape.



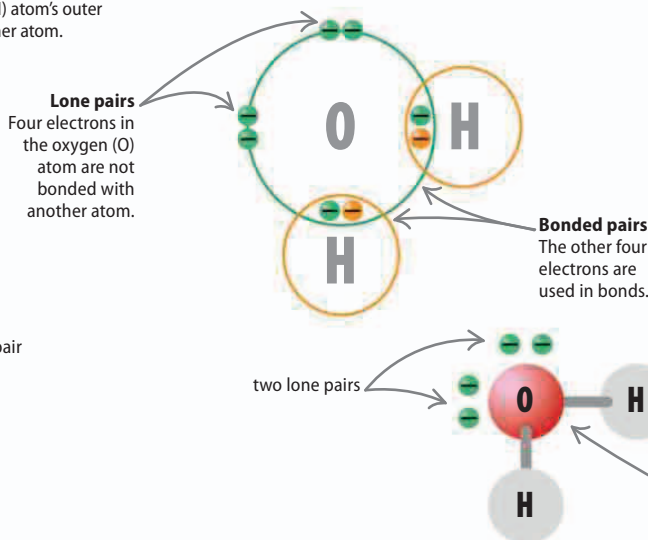
◁ Methane

In a methane molecule there is no lone pair, so the four bonds are repelled equally. This means that the hydrogen atoms are positioned equally around the carbon atom to create a regular shape called a tetrahedron.



△ Ammonia

The lone pair on nitrogen repels the three bonds so they are pushed closer together.



△ Water

In a water molecule, two lone pairs of electrons push the molecule into a V shape.

Intermolecular forces

Most simple covalent compounds are gases, because their molecules stay separate from each other in normal conditions. However, in a liquid or solid, weak intermolecular forces act between the molecules to hold them together. A common type is the dipole-dipole interaction, which occurs between dipoles (molecules that have one negatively charged side and one positively charged side). A negative end of a dipole on one molecule will then attract a positive end of dipole on a neighbouring molecule, holding the two molecules together.

Negative pole

Electrons gather on the far side of the chlorine atom to form the negative end of the dipole.

Positive pole

Because the electrons have moved away from the hydrogen atom, it becomes the positive side of the dipole.

